

GROUP PROJECT TASK

ISP565/ITS665 DATA MINING

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DESCRIPTION OF TASK

A1. Data Acquisition and Data Understanding

1. Search and select a dataset depending on your interests. It should contain enough instances (**at least 3000**), attributes with at least **20 attributes up to 30 attributes including the class label**, should contain a good **mix of numeric and nominal** attributes and if possible, the dataset has **some missing values**. (If there are no missing values, than you need to perform other relevant processes).
2. Describe about your project problem, data and the source of dataset.
3. Each group is required to develop **one** method only – classification. (If your group is interested to do clustering, please refer to your lecturer).

Task in A2-A3-A4 is for data understanding, preparation and reduction.

Meanwhile, task in B is for model development and evaluation.

Use WEKA tool to perform each of the task. You might need other tool such as Excel or other.

The relevant attributes should be within 10-15 attributes.

A2. Data preparation – selection and cleaning

Data selection refers to choose the most relevant attributes or samples (**at minimum of 10 attributes**). Meanwhile, data cleaning can be accomplished by applying filters to the data in the Preprocess tab.

1. Start with the Preprocess tab. Study **numeric attributes**. Give the mean, min & max and its standard deviation.
2. Study the **nominal attributes** and report the values of each attribute and the count of each.
3. **Identify the attributes with missing/noise values**. Remove the missing/noise values with the method of your choice using WEKA, explaining which filter you are using and why you make this choice.
4. **Identify the attributes with outliers**. Investigate the method for **detecting outliers**. Are there any outliers in this dataset, and if yes, describe how you deal with these outliers.
5. Save the cleaned dataset into **file-cleaned.arff**. Show several samples in the dataset, at least the first 20 rows of this dataset – with the columns.

A3. Data preparation – transformation

Among the different data transformation techniques, explore those available through the WEKA filters in the Preprocess tab. Study the following data transformation and report which you have applied:

1. **Perform normalization when necessary**. Explain which filter that you applied; Min-max normalization, Z-score normalization or Decimal normalization and provide detailed information the method of your choice – state which one you choose and why.
*You may not require normalizing all attributes. Explain why you do the normalization on the attributes.
2. **Perform discretization when necessary**. Which attributes and how many bins have you implemented? Explain.
3. **Perform attribute construction when necessary** – for example adding an attribute representing the sum of two other ones. Which Weka filter permits to do this? Show the attribute if you applied it.
4. Perform other specific processes when necessary and explain.
5. Save the prepared dataset into **file_processed.arff**. Show the samples, at least the first 20 rows of this dataset – with all columns.

A4. Data preparation- reduction

This task **should** be done after you have run the model in part B using **relevant attributes from the previous steps in A1-A2-A3**.

Usually, data mining datasets are too large to process directly. Reduction can be done on the attributes (Select attribute) and also on the samples (Sampling). In this project, you must apply **Select Attribute**.

Reduce the **dataset** through **Select Attribute**, using suitable method.

1. Explain your reduced dataset.
2. Compare the results based on several set of features, preferably with more than one different set.
3. Save reduced dataset into **file_reduced.arff**, and paste a screenshot showing at least the first 20 rows of this dataset – with all columns.

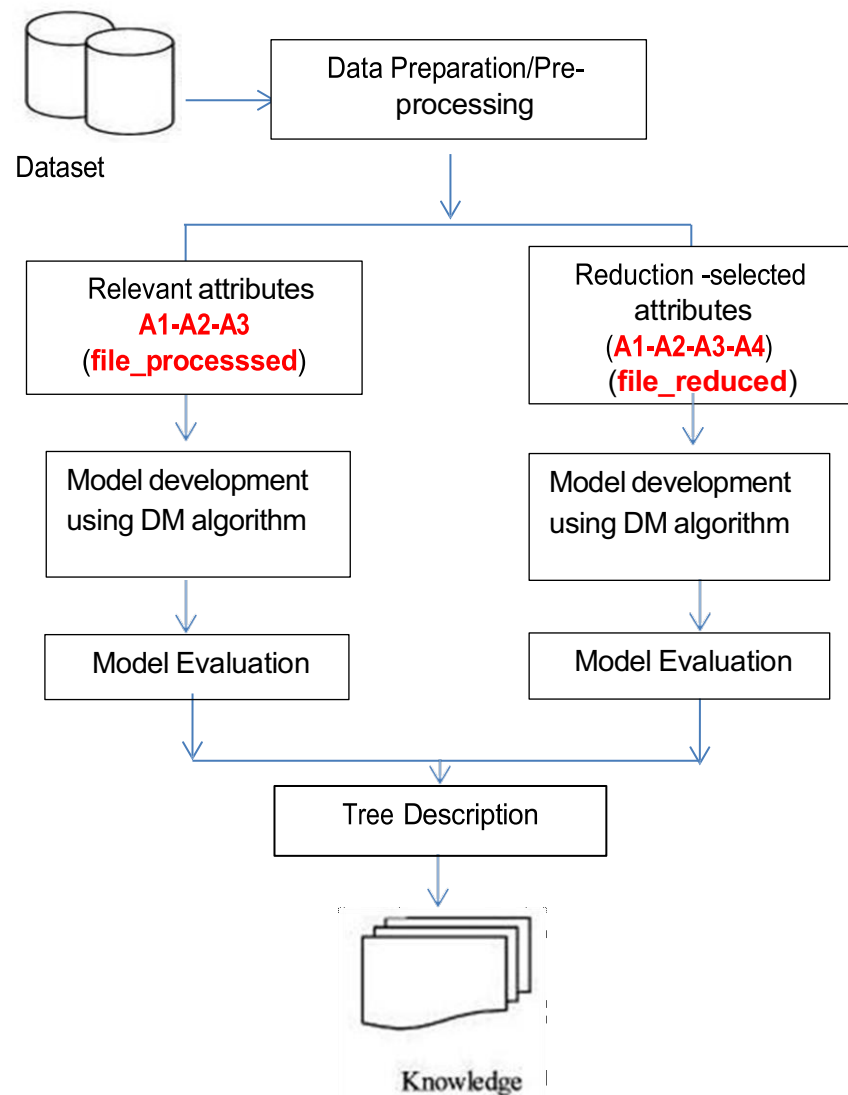
B. Model Development and Evaluation

By default, each group requires to develop classification model. Apply **an algorithm** under selected study using your dataset. Present the outcome of the project. Each member has to elaborate his/her role/contribution for groupwork.

(If your group is interested to do clustering, please refer to your lecturer).

METHOD	ACTIVITIES AND EVALUATION
CLASSIFICATION	1. Perform all tasks in steps A1-A3 .
	2. Testing your results over the training data on the following options; i. Cross validation with different folds, $k = 10$ and $k = 20$. ii. Percentage split (70:30); where 70 is the percentage of training dataset and (80:20); where 80 is the percentage of training dataset. Discuss every result.
	3. Generate the tree visualizer of best result from cross validation and percentage split.
	4. Apply reduction steps in A4 . Report the reduction method that you have applied.
	5. Repeat step 2-3 on the reduction datasets.
	6. Compare the evaluation results of processed dataset and after the dataset is reduced using graph (Excel). Explain your results with the help of the graph (Excel).
OPTIONAL: CLUSTERING	1. Perform all tasks in steps A1-A3 .
	2. Solve the problem using the clustering algorithm in WEKA. Evaluate three different numbers of clusters by investigating the errors (says, $k = \{3, 4, 5, 6, \dots\}$) with following options; i. Euclidean distance ii. Manhattan distance Can you find the best number of clusters? Discuss.
	3. Visualize the clusters using appropriate scatter plots and graphs. Explain with the help of graphs / Excel.
	4. Apply reduction steps in A4 . Report the reduction method that you have applied.
	5. Repeat step 1-3 on the reduction datasets.
	6. Compare result between processed and reduced datasets. Explain the differences of generated clusters, with the help of graphs.

FLOW OF THE TASKS IN THE PROJECT



About the group project task

1. This is a group task of **5 members** to perform data mining project, present the outcome and submit the slides and datasets/models.
2. Finding the right data would be the most tedious task, please spend time in this task. Confirm your datasets with lecturer. **No group can use the same dataset or existing data in WEKA folder**, following the basis of FCFS. The delay in finding the dataset and getting the approval will delay your work.

Useful link to data repositories containing multiple datasets to choose from:

- <http://www.kaggle.com/>
- UCI ML Data Repository <http://archive.ics.uci.edu/ml/datasets.html> (use the recent from 2015 onwards)
- Any related portals

GROUP PROJECT PRESENTATION AND SUBMISSION

Presentation

1. List all the members and picture of each member (with name) in the slide.
2. The presentation is suggested in week 13-14, through live presentation (online or classroom).
3. Each group is given a maximum of 30 minutes slots:
 - a. Presentation should be within maximum 15 minutes time, best is in 10 minutes. Avoid putting many text in the slides OR avoid to read each of the word. Describe briefly.
 - b. 10 – 15 minutes for Q&A. Each group member has to be presence in the class or during the slot with **CAMERA ON** and answer the question properly.
4. During the session, make sure you have your slide, WEKA and the datasets/models on the desktop sharing.

Submission

1. Presentation slide - contains all the results as required, list the group members and pictures in the first slide. Reduce repetition or share common processes.
2. The complete dataset: the original dataset, in .CSV/ARFF format including preprocessed dataset, cleaned, normalized, processed, reduced, train and test datasets etc.
3. Model of the experiments (in WEKA format).
4. Articles for the project.
 - Upload in the MSTEAMS according to group, with naming format:
 - groupID_datafilename_leadername
 - e.g. CS2434A_soccerdata_ali

PROJECT TIMELINE

	2025							
ACTIVITIES	WEEK 7	WEEK 8	WEEK 9	WEEK 10	WEEK 11	WEEK 12	WEEK 13	WEEK 14
Understanding and Information Gathering								
Data Preparation								
Model Development								
Analysis and Results								
Delivery of Output (Presentation)								

REFERENCE for RUBRIC

DATA PREPARATION	(CLO3-P5 /Digital Skill/ PLO12-eres)	Subattribute	Level	1 - 2 (Very Weak)	3 - 4 (Weak)	5 - 6 (Fair)	7 - 8 (Good)	9-10 (Very Good)
Identifying techniques for data preparation in doing Data Mining project		Problem Identification	Initial phase of study	Not able to identify the techniques for data preparation	Able to partially identify the basic techniques for data preparation (e.g. data collection, data selection)	Able to partially identify the techniques for data preparation with basic explanation	Able to identify the techniques for data preparation with clear explanation	Able to identify and provide explanation of data preparation techniques very clearly and accurately
Analysing the data preparation results with DM technology/tools		Analysis	Initial phase of study	Not able to analyse the data preparation results	Finds difficulty in analysing the data preparation results	Able to organise and analyse the data preparation results but does not clearly analyse the factors that contribute to the problem	Able to organise and analyse the data preparation results and clearly analyse the factors that contribute to the problem	Able to organise and analyse the data preparation results and very clearly analyse the factors that contribute to the problem
MODEL DEVELOPMENT (A1-A2-A3-A4)	(CLO3-P5 / PLO12)	Subattribute	Level	1 - 2 (Very Weak)	3 - 4 (Weak)	5 - 6 (Fair)	7 - 8 (Good)	9-10 (Very Good)
Applying DM algorithms for model building in the selected dataset (A1-A2-A3)		Application	Middle phase of the project	Not able to develop the data mining model	Finds difficulty in developing the data mining models	Able to develop the data mining models but does not clearly explain the factors/parameters that involved	Able to develop the data mining models and clearly explain the factors/parameters that involved	Able to develop the data mining models and very clearly explain the factors/parameters that involved
Applying DM algorithms for model building in the reduced dataset (A1-A2-A3-A4)		Application	Middle phase of the project	Not able to develop the data mining model on the reduced dataset	Finds difficulty in developing the data mining models with the reduction dataset	Able to develop the data mining models on the reduction dataset but does not clearly explain the factors/parameters that involved	Able to develop the data mining models on the reduction dataset and clearly explain the factors/parameters that involved	Able to develop the data mining models on the reduction dataset and very clearly explain the factors/parameters that involved
PRESENTATION	(CLO3-P5 / PLO12)	Subattribute	Level	1 - 2 (Very Weak)	3 - 4 (Weak)	5 - 6 (Fair)	7 - 8 (Good)	9-10 (Very Good)
Presentation (preparation-model development-evaluation)		Presentation	Final phase of the project	Not able to present the results of model evaluation (preparation-model development-evaluation-accuracy, precision, recall etc)	Able to present the results of model evaluation (preparation-model development-evaluation-accuracy, precision, recall etc)	Able to present the data mining models but does not clearly present the comparison of the models	Able to present and visualize (e.g. Decision Tree model) the data mining models and clearly present the comparison of the models	Able to present and visualize (e.g. Decision Tree model) the data mining models and clearly present the comparison of the models

*Subject to change

GROUP INFORMATION

Find the form in MSTEAM.

Group number	Group Members	Student ID	Project Title	Dataset Link	
CS2594A-1					
CS2594A-2					
CS2594A-3					
CS2594B-1					
CS2594B-2					
CS2594C-1					
CS2594C-1					